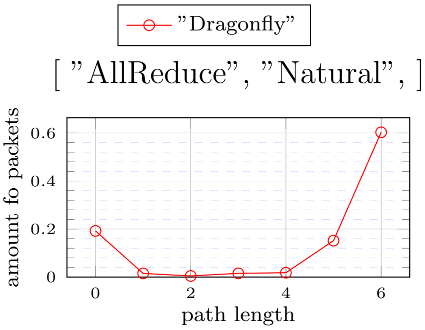

Figure 1: X: ["AllReduce", "Natural",]

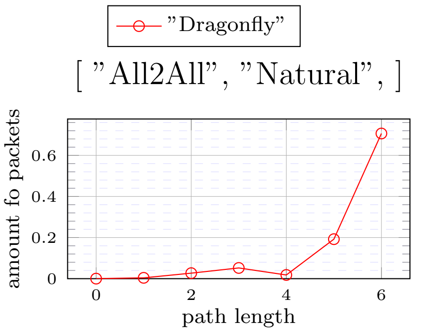

Figure 2: X: ["All2All", "Natural",]

The following versions used in the simulations.

- heads/master-dirty-ce88a576295ce68f40736599a9e3e04f24f38801(0.6.3)
- heads/master-dirty-65e4f7db651d154c0abb3e3d0a691be7d6d92d82(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 10000000000000000,
  general_frequency_divisor: 2,
  topology: topologia![
    Dragonfly{ global_ports_per_router: 8, servers_per_router: 8, global_arrangement: Palmtree, group_size: 16, number_of_groups: 129, legend_name: "Dragonfly" }],
  traffic: ![
    TrafficMap{
      tasks: 16512,
      map: ![
        Composition{
          patterns: [
            Identity{ legend_name: "Natural" },
            CartesianEmbedding{
              source_sides: [ 16384 ],
              destination_sides: [ 16512 ]}],
          middle_sizes: [ 16384, 16512 ],
          legend_name: "Natural"}],
      application: ![
        AllReduce{ tasks: 16384, data_size: 16384, legend_name: "AllReduce" }],
      legend_name: "AllReduce"},
    TrafficMap{
      tasks: 16512,
      map: ![
        Identity{ legend_name: "Natural" }],
      application: ![
        All2All{ tasks: 16512, data_size: 132096 }],
      legend_name: "All2All"}],
  maximum_packet_size: 16,
  router: InputOutput{
    virtual_channels: topologia![ 4 ],
    virtual_channel_policies: topologia![
      [
        MapLabel{
          label_to_policy: [
            Chain{
              policies: [
                OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true, use_neighbour_space: true, aggregate: false }]},
            Chain{
              policies: [
                AverageOccupancyFunction{
                  label_coefficient: 0,
                  occupancy_coefficient: 2,
                  product_coefficient: 0,
                  constant_coefficient: 16,
                  use_internal_space: true,
                  use_neighbour_space: true,
                  exclude_minimal_ports: true,
                  virtual_channels: [ 0 ]}]}]},
        LowestLabel,
        EnforceFlowControl,
        Random]],
    allocator: Random,
    buffer_size: 128,
    bubble: false,
    flit_size: 16,
    intransit_priority: false,
    allow_request_busy_port: true,
    output_buffer_size: 64,
    crossbar_frequency_divisor: 1,
    crossbar_delay: 2},
  routing: topologia![
    Sum{
      policy: TryBoth,
      first_routing: ChannelMap{
        routing: AscendantChannelsWithLinkClass{
          routing: WeighedShortest{
            class_weight: [ 1, 100 ]},
          bases: [ 1, 1 ]},
        map: [
          [ 0 ],
          [ 1 ]]},
      second_routing: Valiant4Dragonfly{
        first: ChannelMap{
          routing: AscendantChannelsWithLinkClass{
            routing: WeighedShortest{
              class_weight: [ 1, 100 ]},
            bases: [ 1, 1 ]},
          map: [
            [ 0 ],
            [ 1 ]]},
        second: ChannelMap{
          routing: AscendantChannelsWithLinkClass{
            routing: WeighedShortest{
              class_weight: [ 1, 100 ]},
            bases: [ 1, 1 ]},
          map: [
            [ 2 ],
            [ 3 ]]}},
      first_allowed_virtual_channels: [ 0, 1, 2, 3 ],
      second_allowed_virtual_channels: [ 0, 1, 2, 3 ],
      second_extra_label: 1,
      legend_name: "Ugal"}],
  link_classes: [
    LinkClass{ delay: 2 },
```

LinkClass { **delay**: 2 },
LinkClass { **delay**: 2 }],
launch_configurations: [
Slurm { **job_pack_size**: 1, **time**: "3-00:00:00" }]}